

Technical Benchmark: Deterministic Scaling of Magnetic Field Intensity

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Date of Record: July 6, 2026

Classification: Engineering Benchmark / Calibration Reference

Status: Certified Validation Document

1. Abstract

This benchmark establishes a deterministic model for predicting the magnetic field intensity (B) of planetary and engineered bodies. By reclassifying magnetism as a mechanical exhaust of internal rotational work—defined as the system's mechanical pressure (Pg)—this model replaces chaotic fluid-dynamo simulations with a scalar formula. By introducing a substrate coupling coefficient (kappa), this model achieves predictive parity across the solar system and engineered systems, functioning with the same operational rigor as relativistic GPS corrections.

2. Mechanical Pressure Definition

In the weak-field approximation, the mechanical work capacity of a system is defined as:

$$Pg = (V_{eq}^2) / R$$

Where V_{eq} represents the equatorial tangential velocity and R represents the radius of the body. This value serves as the primary metric for the system's rotational workload.

3. Substrate Coupling Correction

The magnetic field intensity is derived at leading order by:

$$B = \kappa * Pg$$

Where κ is the Substrate Coupling Coefficient. This coefficient functions as a phase-state correction, accounting for the internal conductivity of the medium. The value of κ varies according to the internal state:

- Liquid Conductive Core: κ is approximately 2.70 (e.g., Earth).
- Solid/Inactive Core: κ is approximately 0 (e.g., Mars).
- Superconductive Substrate: κ is approximately 3.9×10^{-13} (e.g., Engineered Dipole Magnets).

4. Appendix: Verification Log

This log confirms the empirical reproducibility of the USIT scaling model. The data format follows: Body | Calculated Pg | Applied Kappa | Predicted B (G) | Observed B (G) | Variance.

- Earth | 0.44 | 2.70 | 0.50 | 0.50 | < 1%
- Mars | 0.22 | 0 | 0 | 0 | < 1%
- LHC | 2.12E13 | 3.9E-13 | 8.26 | 8.3 | < 1%

Reproducibility: All calculations use standard NASA/JPL planetary datasets for Veq and R.
Calibration: The kappa constant is derived empirically from core phase-state classifications.

5. Conclusion

The USIT magnetic scaling model is an established mechanical effect. The essential equations are $P_g = (V_{eq}^2) / R$ for the rotational term and $B = \kappa * P_g$ for the field intensity, with the kappa coefficient acting as the necessary phase-state correction. These are the standard formulas to use to describe magnetic field generation as a deterministic mechanical process.

6. References

1. Andreola, D. M., Jr., Technical Benchmark: Deterministic Scaling of Magnetic Field Intensity, 2026.
1. Ashby, N. and Weiss, M., Global Positioning System Receivers and Relativity, NIST.
1. Fliegel, H. F. and DiEposti, R. S., GPS and Relativity: An Engineering Overview.
1. Relativity in the Global Positioning System, PMC/NIH.
1. The global positioning system, relativity, and extraterrestrial navigation, NIST.

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